Introduction to Probability and Statistics

(EE 354 / CE 361)

Joint PMFs

1. $\mu_X = E[X] = \sum_x x \cdot p_X(x)$ *

2. Expected Value Rule

$Y = g(X)$

$$\mu_Y = E[Y] = E[g(X)] = \sum_x g(x) p_X(x)$$

3. $\text{var}(X) = E[(X - E[X])^2]$

$f(x) = (X - E[X])^2$

$$\text{var}(X) = \sum_x (x - E[X]^*)^2 p_X(x)$$

↳ Constant $= \mu_X$

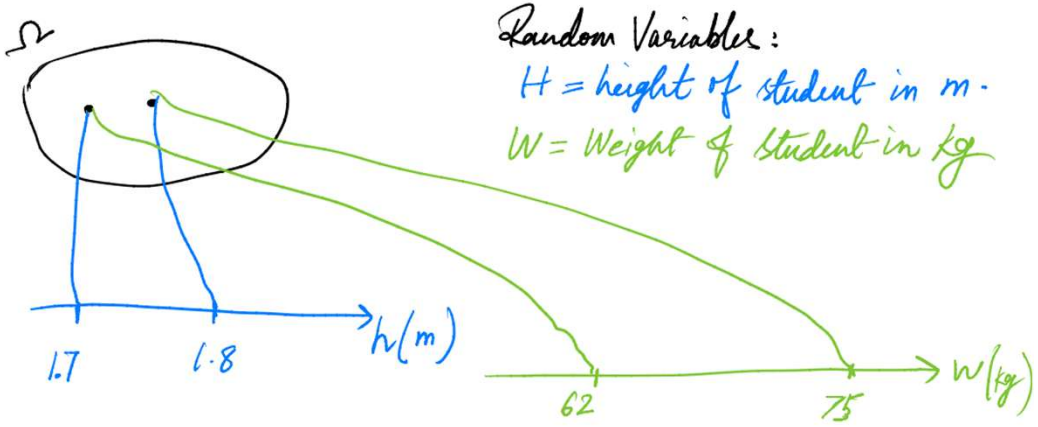
Today's Agenda

- Joint PMFs
  - Definition/Intuition
  - Properties
  - Extension to more than Two Random Variables
  - Mean
  - Mean of Binomial Distribution

Multiple Random Variables on the same Sample Space

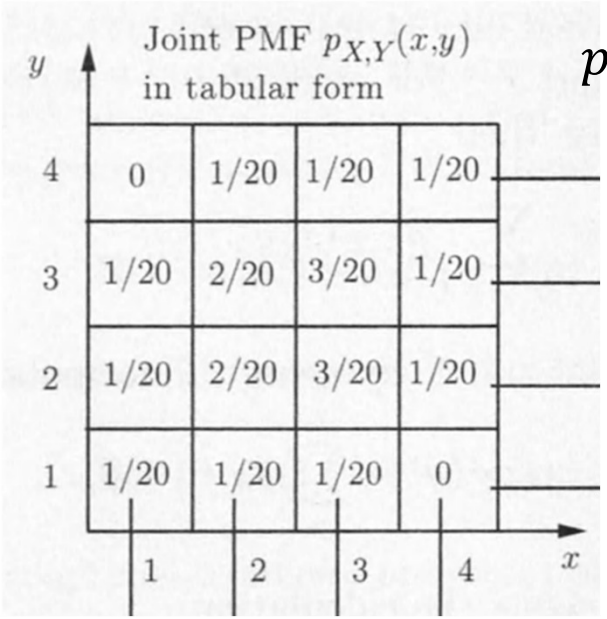
- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class

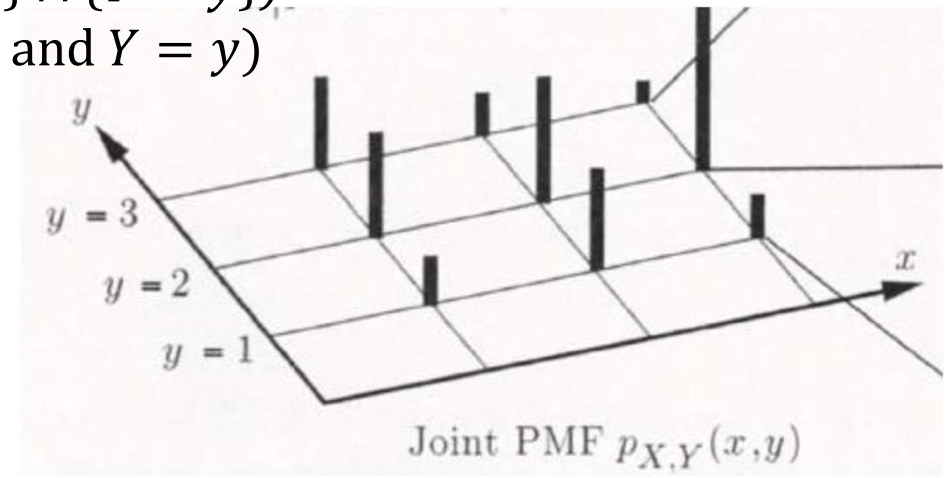


Multiple Random Variables and Joint PMF

- We can extend the concepts of PMF and expectation developed so far for a single RV to multiple RVs
- Consider two discrete random variables X and Y associated with the same experiment.
  - The probabilities of the values that X and Y can take are captured by the joint PMF of X and Y, denoted by $p_{X,Y}$
  - In particular, if (x, y) is a pair of possible values of X and Y, the probability mass of (x, y) is the probability of the event $\{X = x, Y = y\}$



$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y)$$
$$= \mathbf{P}(\{X = x\} \cap \{Y = y\})$$
$$= \mathbf{P}(X = x \text{ and } Y = y)$$



Properties of Joint PMF

$$\boxed{\sum_x \sum_y p_{X,Y}(x,y) = 1}$$

- We can calculate the PMFs of X and Y from Joint PMFs:

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

The formula for $p_X(x)$ can be verified using the calculation

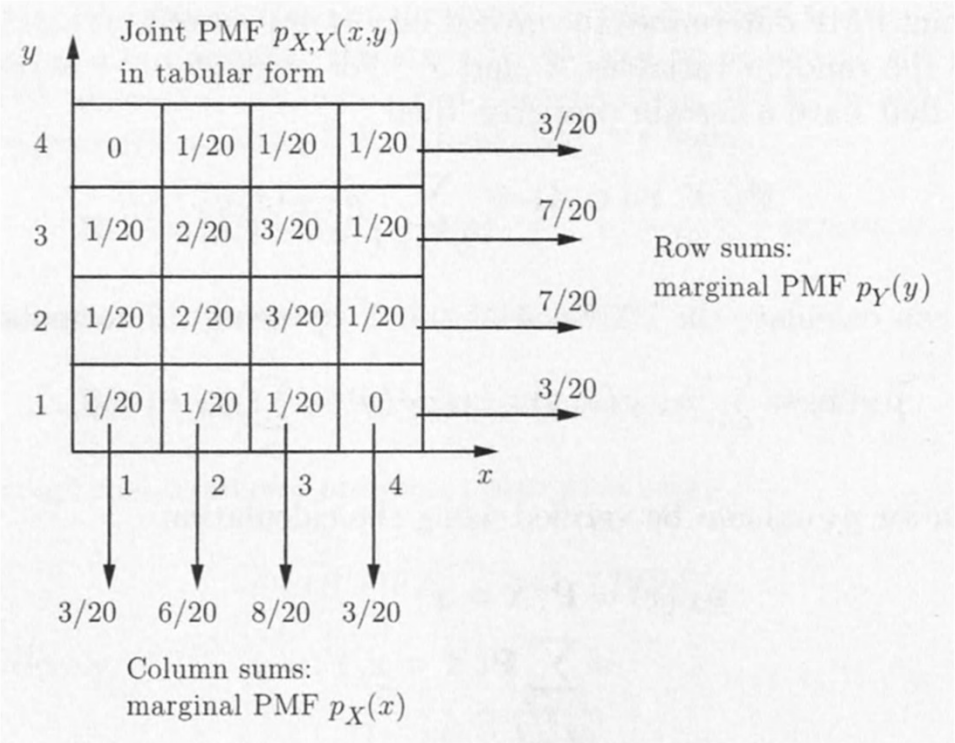
$$\begin{aligned} p_X(x) &= \mathbf{P}(X = x) \\ &= \sum_y \mathbf{P}(X = x, Y = y) \\ &= \sum_y p_{X,Y}(x,y), \end{aligned}$$

where the second equality follows by noting that the event $\{X = x\}$ is the union of the disjoint events $\{X = x, Y = y\}$ as y ranges over all the different values of Y. The formula for $p_Y(y)$ is verified similarly. We sometimes refer to p_X and p_Y as the **marginal** PMFs, to distinguish them from the joint PMF.

Properties of Joint PMF

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$



Joint PMF: More than Two Random Variables

- The joint PMF of three random variables X, Y, and Z is defined as:

$$p_{X,Y,Z}(x, y, z) = \mathbf{P}(X = x, Y = y, Z = z).$$

- Also,

$$\sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$$

- Corresponding marginal PMFs can also be analogously obtained:

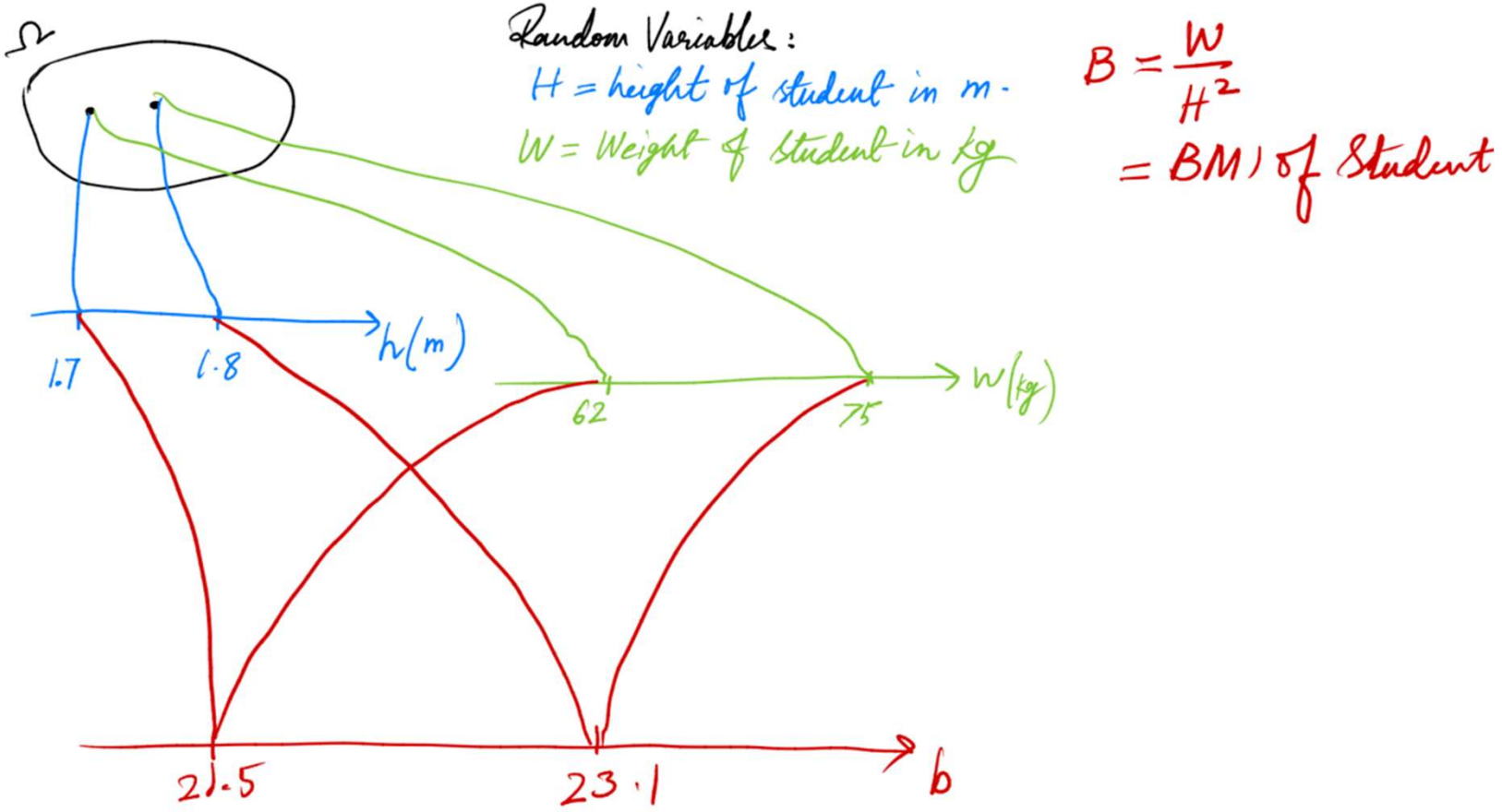
$$p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z),$$

and

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x, y, z).$$

Remember: Functions of Random Variables

- A function of one or several random variables is also a random variable.
- Meaning of $X + Y$: Our RV takes on value $(x + y)$ when X takes on value x and Y takes on value y



Functions of Random Variables: Context/Utilization

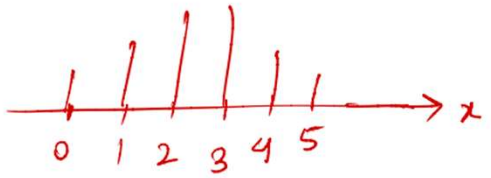
X : No. of red signals out of 5 on my commute

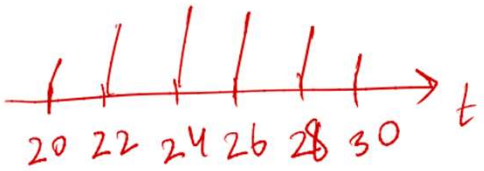
T : Commute time

$$T = 20 + 2X$$

$$E[X] = \sum_x x\, p_X(x)$$

$$\mu_X \,,\ \mu_T = 20 + 2\mu_X$$





Functions of Multiple Random Variables

- We can extend the concepts of PMF and expectation developed so far for functions of a single RV to functions of multiple RVs.

- If $Z = g(X, Y)$ is a function of random variable X and Y, then Z is also a random variable and its PMF can be calculated from the joint PMF $p_{X,Y}$ as follows:

$$p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{\{(x,y) \mid g(x,y)=z\}} p_{X,Y}(x, y).$$

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- Expected Value Rule:

$$\mathbf{E}\big[g(X, Y)\big] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y).$$

- In the special case where g is linear and of the form $aX + bY + c$, where a, b, and c are given scalers, we have:

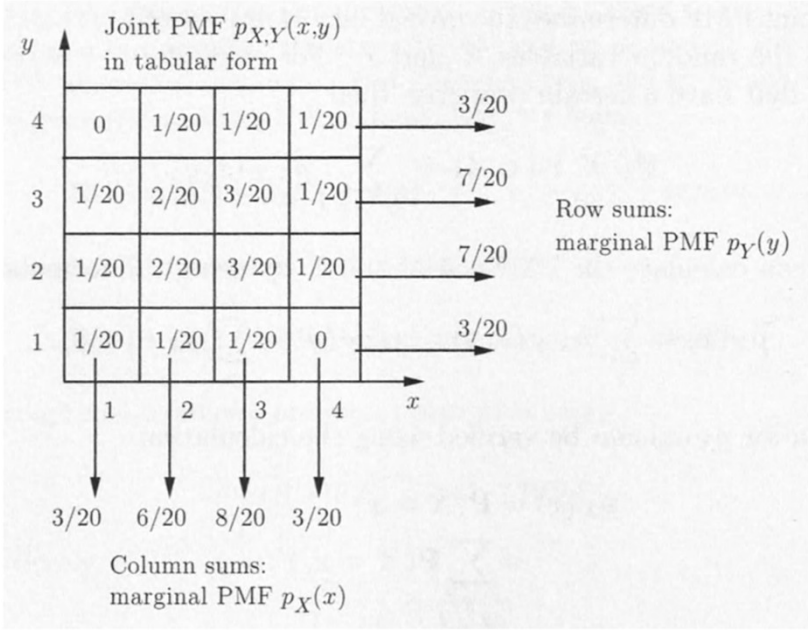
$$\mathbf{E}[aX + bY + c] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c.$$

Example: Functions of Multiple Random Variables

- Consider our previous example:
- Consider a new random variable Z:

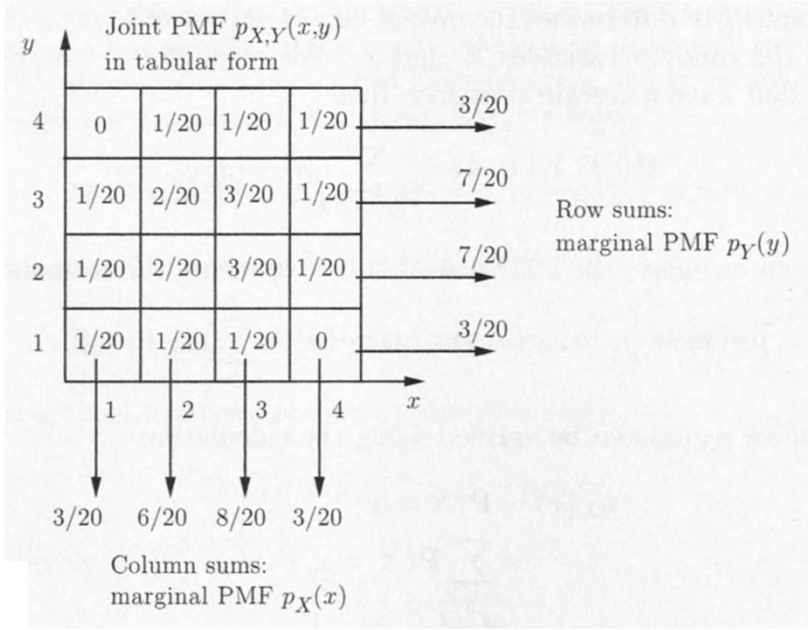
$$Z = X + 2Y$$

- What is the PMF of Z?



Example: Functions of Multiple Random Variables

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- Consider a new random variable Z:

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- What is the PMF of Z?

$$p_Z(z) = \sum_{\{(x,y)\,|\,x+2y=z\}} p_{X,Y}(x,y),$$

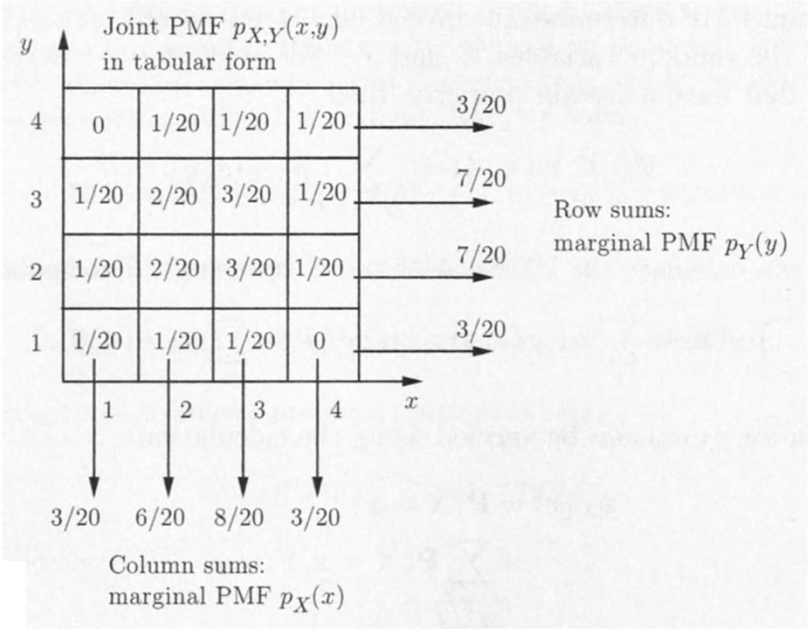
$$p_Z(3) = \frac{1}{20},\quad p_Z(4) = \frac{1}{20},\quad p_Z(5) = \frac{2}{20},\quad p_Z(6) = \frac{2}{20},\quad p_Z(7) = \frac{4}{20},$$

$$p_Z(8) = \frac{3}{20},\quad p_Z(9) = \frac{3}{20},\quad p_Z(10) = \frac{2}{20}.\quad p_Z(11) = \frac{1}{20},\quad p_Z(12) = \frac{1}{20}.$$

$$\begin{aligned}\mathbf{E}[Z] &= \sum z p_Z(z) \\ &= 3\cdot\frac{1}{20} + 4\cdot\frac{1}{20} + 5\cdot\frac{2}{20} + 6\cdot\frac{2}{20} + 7\cdot\frac{4}{20} \\ &\quad + 8\cdot\frac{3}{20} + 9\cdot\frac{3}{20} + 10\cdot\frac{2}{20} + 11\cdot\frac{1}{20} + 12\cdot\frac{1}{20} \\ &= 7.55.\end{aligned}$$

Example: Functions of Multiple Random Variables

- Consider our previous example:



- Consider a new random variable Z:

$$Z = X + 2Y$$

- What is the PMF of Z?

$$p_Z(z) = \sum_{\{(x,y)\,|\,x+2y=z\}} p_{X,Y}(x,y),$$

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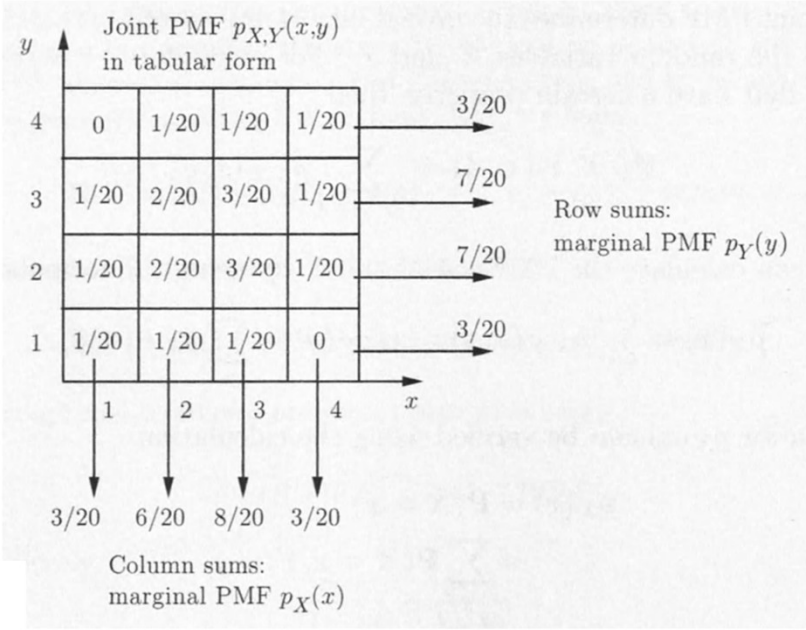
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- We can refer to the fact that Z is a **linear function** of X and Y

$$\mathbf{E}[Z] = \mathbf{E}[X] + 2\mathbf{E}[Y].$$

$$\mathbf{E}[X] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{6}{20} + 3 \cdot \frac{8}{20} + 4 \cdot \frac{3}{20} = \frac{51}{20},$$

$$\mathbf{E}[Y] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{7}{20} + 3 \cdot \frac{7}{20} + 4 \cdot \frac{3}{20} = \frac{50}{20},$$

$$\mathbf{E}[Z] = \frac{51}{20} + 2 \cdot \frac{50}{20} = 7.55.$$

Extensions to More than Two Random Variables

if g is linear and has the form $aX + bY + cZ + d$, then

$$\mathbf{E}[aX + bY + cZ + d] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c\mathbf{E}[Z] + d.$$

Furthermore, there are obvious generalizations of the above to more than three random variables. For example, for any random variables $X_1, X_2, \ldots, X_n$ and any scalars $a_1, a_2 \ldots, a_n$, we have

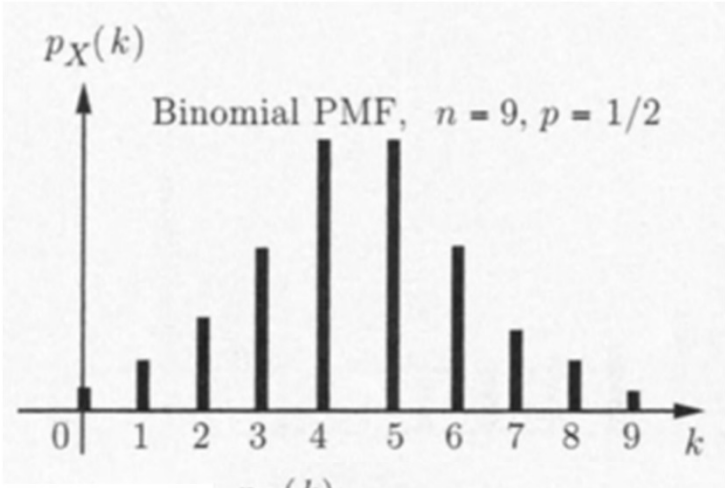
$$\mathbf{E}[a_1X_1 + a_2X_2 + \cdots + a_nX_n] = a_1\mathbf{E}[X_1] + a_2\mathbf{E}[X_2] + \cdots + a_n\mathbf{E}[X_n].$$

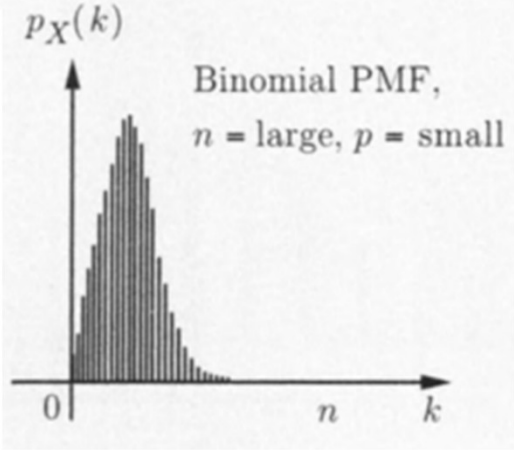
- Example

  $\mathbf{E}[2X + 3Y - Z] = \mathbf{E}[2X] + \mathbf{E}[3Y] - \mathbf{E}[Z] = 2\mathbf{E}[X] + 3\mathbf{E}[Y] - \mathbf{E}[Z]$

Binomial Random Variable

- Experiment
  - n independent tosses of a coin where at each toss, the coin comes up a head with probability p and a tail with probability $(1 - p)$.
- Sample Space
  - Set of Head (H) and Tail (T) sequences of length n
- Random Variable
  - The number of Heads in the n-toss sequence
- PMF
  - $\boldsymbol{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \ldots, n}$
- Model of:
  - Number of successes in a given number of independent trials
- Parameters: $\boldsymbol{n \in \mathbb{Z}^{++}}$, $\boldsymbol{p \in [0, 1]}$





Mean of a Binomial Random Variable

- Random Variable
  - The number of Heads in the n-toss sequence
- PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \dots, n$$

- Mean

$$\mathbf{E}[X] = \sum_{k=0}^{n} k p_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

  - An alternative approach:

Lets consider n random variables $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

$X_1, X_2, \dots, X_n$ are Bernoulli random variables with common mean p
Their sum

$$X = X_1 + X_2 + \cdots + X_n$$

is a binomial random variable with parameters n and p.

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n]$$

$$\mathbf{E}[X] = np$$

Example: Mean of a Binomial Random Variable

- Example
  - Your probability class has 60 students and each student has probability $\frac{1}{3}$ of getting an A, independent of any other student. Consider the RV:

    X = The number of students that get an A
  - What is the mean of X?

Example: Mean of a Binomial Random Variable

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Note that X is a binomial RV.

$$\mathbf{E}[X] = np$$

$$\mathbf{E}[X] = 60\left(\frac{1}{3}\right) = 20$$

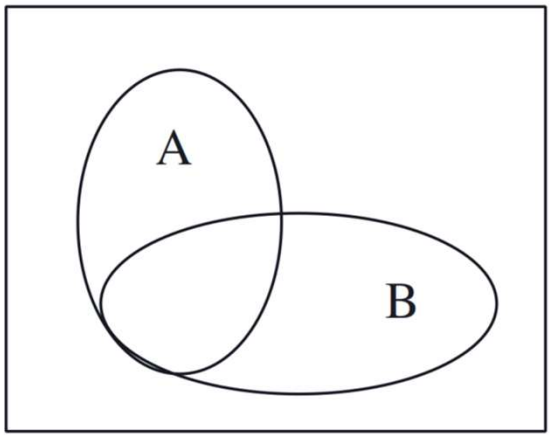
Recall: Conditional Probability

$\mathbf{P}(A \mid B) =$ probability of A, given that B occurred

- B is our new universe

Definition: Assuming $\mathbf{P}(B) \neq 0$,

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$



Conditioning a Random Variable on an Event

- Conditional PMF
  - The conditional PMF of a random variable X, conditioned on a particular event A with $P(A) > 0$, is defined by

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}.$$

Remember:

$$p_X(x) = \mathbf{P}(X = x)$$

  - Similar to the unconditional case

$$\sum_x p_{X|A}(x) = 1,$$

$$\sum_x p_X(x) = 1$$

- Conditional Expectation

$$\mathbf{E}[X] = \sum_x x p_X(x) \longrightarrow \mathbf{E}[X|A] = \sum_x x p_{X|A}(x)$$

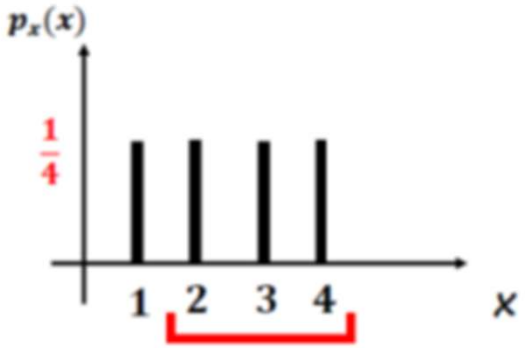
$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x) \longrightarrow \mathbf{E}[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

Example: Conditioning a Random Variable on an Event

- Let X be a Random variable such that

$$p_X(x) = \begin{cases} \frac{1}{4}, & x = 1,2,3,4 \\ 0, & \text{otherwise} \end{cases}$$

- Let $A = \{X \geq 2\}$.
  - What is $p_{x|A}(x)$?

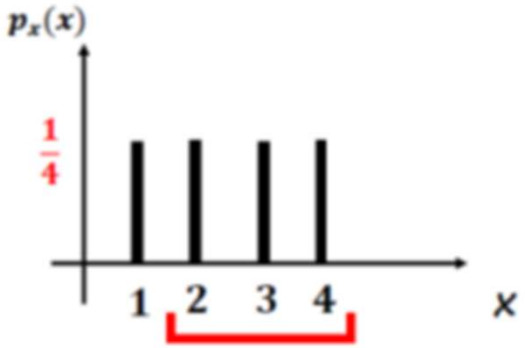


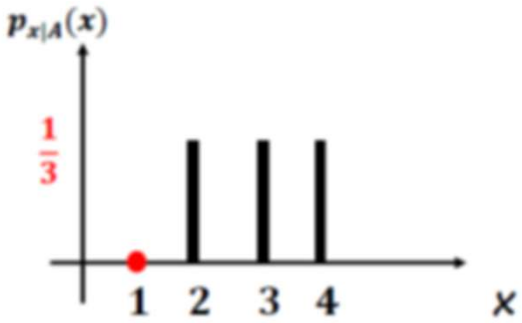
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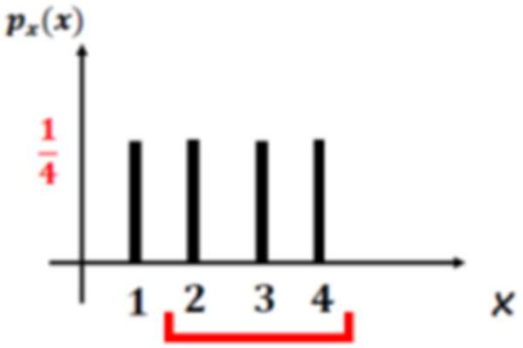
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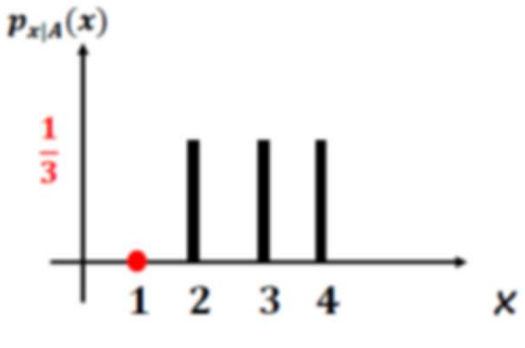




Example: Conditioning a Random Variable on an Event

- Let $A = \{X \geq 2\}$.





$\mathbf{E}[X] =$

$\mathbf{E}[X|A] =$

$\mathrm{var}[X] =$

$\mathrm{var}[X|A] =$